

Data Science Toolbox Portfolio Questions

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Block 4 Unsupervised learning - Latent Structures

Portfolio 04

Portfolio Question 4.1.P - Constrained Optimisation

- How is Constrained Optimisation related to Eigenvalues and Eigenvectors?
 - Is this trick relevant to wider machine learning?

Portfolio Question 4.2.P - Mixture Modelling

- “Gaussian Mixture Modelling” theoretically dominates “soft K-means”.
 - Explain Gaussian Mixture modelling and how it relates to K-means as described in the lecture.
 - Is this claim true? If so, why would we ever use K-means?

Portfolio Question 04.3.P - Dimensionality

- How would you go about making a real classifier from a clustering model, that can be deployed on a training dataset?
 - Could you use the classifier to learn the “correct” number of dimensions? How?
 - Is the dimension of all embedding methods the same?
 - Is dimension as chosen above the only concept of dimension?
 - Back up your exploration with a demonstration.